ANSWER KEY — Integration Lesson 17 Test: Substitution I (Linear Insides)

For the grader · full working shown · give credit for correct method with small arithmetic slips · missing + C costs half · the classic error is forgetting to divide by the inside's slope

1. Find $\int(x+3)^4\,dx$.

> **Work:** inside $x+3$ has slope 1. Integrate as if the inside were one letter: $(x+3)^5/5$, then divide by 1 (no change). Check: $5(x+3)^4\cdot 1/5 = (x+3)^4$ ✓. **Answer: $(x+3)^5/5 + C$.**

2. Find $\int(2x+5)^2\,dx$, and check by differentiating.

> **Work:** as-if-one-letter: $(2x+5)^3/3$; inside slope is 2, so divide by 2 → $(2x+5)^3/6$. Check: $3(2x+5)^2\cdot 2/6 = (2x+5)^2$ ✓. **Answer: $(2x+5)^3/6 + C$.**

3. Find $\int e^{(3x)}\,dx$.

> **Work:** as-if-one-letter: $e^{(3x)}$; divide by the slope 3. Check: $(e^{(3x)}/3)' = 3e^{(3x)}/3 = e^{(3x)}$ ✓. **Answer: $e^{(3x)}/3 + C$.**

4. Find $\int\cos(4x)\,dx$.

> **Work:** $\int\cos$ gives sin (no minus); divide by the slope 4: $\sin(4x)/4$. Check: $4\cos(4x)/4 = \cos(4x)$ ✓. **Answer: $\sin(4x)/4 + C$.**

5. Find $\int\sin(3x)\,dx$.

> **Work:** $\int\sin$ gives $-\cos$ (the Lesson 16 minus); divide by the slope 3: $-\cos(3x)/3$. Check: $(-\cos(3x)/3)' = -(-\sin(3x))\cdot 3/3 = \sin(3x)$ ✓. **Answer: $-\cos(3x)/3 + C$.** (Watch for $+\cos(3x)/3$ or $-\cos(3x)$ without the /3.)

6. A student writes $\int e^{(5x)}\,dx = e^{(5x)} + C$. Show the mistake, then correct it.

> **Work:** differentiate the claim: $(e^{(5x)})' = 5e^{(5x)}$ — five times too big, because the chain rule multiplies by the inside's slope 5. Fix: divide by 5. Verify: $(e^{(5x)}/5)' = 5e^{(5x)}/5 = e^{(5x)}$ ✓. **Answer: the student forgot to divide by 5; correct answer $e^{(5x)}/5 + C$.**

7. Compute $\int_0^1 (6x+2)^2\,dx$.

> **Work:** antiderivative: $(6x+2)^3/3 \div 6 = (6x+2)^3/18$. (Check: $3(6x+2)^2\cdot 6/18 = (6x+2)^2$ ✓.) Inside at the limits: $6\cdot 1+2 = 8$ and $6\cdot 0+2 = 2$. FTC: $8^3/18 - 2^3/18 = 512/18 - 8/18 = 504/18 = 28$. **Answer: 28.**

8. Compute $\int_0^\pi \cos(x/2)\,dx$.

> **Work:** inside $x/2$ has slope $½$; dividing by $½$ = multiplying by 2, so the antiderivative is $2\sin(x/2)$. (Check: $2\cos(x/2)\cdot½ = \cos(x/2)$ ✓.) FTC: $[2\sin(x/2)]_0^\pi = 2\sin(\pi/2) - 2\sin 0 = 2\cdot1 - 0 = 2$. **Answer: 2.**

9. Compute $\int_0^\pi \sin(2x)\,dx$, and explain via signed area.

> **Work:** antiderivative: $-\cos(2x)/2$. FTC: $[-\cos(2x)/2]_0^\pi = (-\cos 2\pi)/2 - (-\cos 0)/2 = (-1)/2 - (-1)/2 = -½ + ½ = 0$. Explanation: $\sin(2x)$ runs at double speed, so on $[0, \pi]$ it completes one FULL cycle — a hump above the axis (0 to $\pi/2$) and a matching hump below ($\pi/2$ to π) that cancel exactly. **Answer: 0 — a full cycle's areas above and below the axis cancel.**

10. Challenge: compute $\int_0^1 (e^{4x} + (3x+2)^2)\,dx$ exactly.

> **Work:** term by term. $\int e^{4x}\,dx = e^{4x}/4$; $\int(3x+2)^2\,dx = (3x+2)^3/3 \div 3 = (3x+2)^3/9$. (Checks: $4e^{4x}/4 = e^{4x}$ ✓; $3(3x+2)^2\cdot3/9 = (3x+2)^2$ ✓.) FTC with $F(x) = e^{4x}/4 + (3x+2)^3/9$. Top $x = 1$: $e^4/4 + 5^3/9 = e^4/4 + 125/9$. Bottom $x = 0$: $e^0/4 + 2^3/9 = 1/4 + 8/9$. Subtract: $(e^4 - 1)/4 + 117/9 = (e^4 - 1)/4 + 13$. **Answer: $(e^4 - 1)/4 + 13$ (≈ 26.4).**